THE SECOND LARGEST OUTDEGREE OF CONDITIONED GALTON–WATSON TREES WITH INFINITE VARIANCE

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ABSTRACT. Let ξ be a random variable with values in $\{0,1,2,\dots\}$ and $\mathbb{E}\xi = 1$, let $Y_{(1)} \geq Y_{(2)} \geq \dots$ be the outdegrees, in decreasing order, of a Galton–Watson tree with offspring distribution ξ conditioned to have n vertices, and let $\xi_{(1)} \geq \xi_{(2)} \geq \dots$ be the order statistics of n independent copies of ξ. Janson proved that if ξ has finite variance then, for every fixed j, $Y_{(j)}$ and $\xi_{(j)}$ have asymptotically the same distribution. He asked whether two consequences of this remain true when the variance is infinite: that $Y_{(1)} < h(n)$ with high probability when $n\mathbb{P}(\xi \geq h(n)) \to 0$, and that $Y_{(j)} \geq h(n)$ with high probability for every fixed j when $n\mathbb{P}(\xi \geq h(n)) \to \infty$. Kortchemski and Richier showed that the first can fail. We show that the second can fail as well. We construct an offspring distribution with mean 1 and infinite variance, and integers $n_r \to \infty$ and L_r with $n_r\mathbb{P}(\xi \geq L_r) \geq r$, such that with probability at least $1 - 1/r$ the conditioned tree with n_r vertices has exactly one vertex of outdegree at least L_r, of outdegree $(r + \frac{1}{2})L_r$, while all other vertices have outdegree at most $L_r/(64(\log n_r)^2)$. Along n_r the largest outdegree is of larger order than $\xi_{(1)}$ and the second largest is of smaller order than $\xi_{(2)}$, and each of the six conclusions of Janson's theorem fails. The same holds for the random allocation of $n-1$ balls into n boxes.

1. INTRODUCTION

Let $w = (w_k)_{k\geq 0}$ be a sequence of nonnegative weights with $w_0 > 0$ and $w_k > 0$ for some $k \geq 1$. For integers $m \geq 0$ and $n \geq 1$, the random allocation $B_{m,n} = (Y_1, \dots, Y_n)$ of m balls into n boxes is the random vector in $\mathbb{Z}^n_{\geq 0}$ with

$$\mathbb{P}(B_{m,n} = (y_1, \dots, y_n)) = \frac{1}{Z(m,n)} \prod_{i=1}^{n} w_{y_i}, \qquad y_1 + \dots + y_n = m,$$

where $Z(m,n)$ is the normalizing constant, assumed to be positive. We write $Y_{(1)} \geq Y_{(2)} \geq \dots \geq Y_{(n)}$ for $Y_1, \dots, Y_n$ in decreasing order, and we follow the notation of Janson's survey [2]. When w is a probability distribution, $B_{m,n}$ is the vector of n independent random variables with distribution w conditioned on their sum being m. If moreover $\mathcal{T}_n$ is a Galton–Watson tree with offspring distribution w conditioned to have n vertices, then by [2, Lemma 17.1] $\mathcal{T}_n$ and $B_{n-1,n}$ can be coupled so that the outdegrees of the vertices of $\mathcal{T}_n$, listed in depth-first order, form a cyclic shift of $B_{n-1,n}$. In particular the outdegrees of $\mathcal{T}_n$ in decreasing order have the same distribution as $(Y_{(1)}, \dots, Y_{(n)})$ for $m = n-1$.

Let $\Phi(t) = \sum_k w_k t^k$, let ρ be its radius of convergence, let $\Psi(t) = t\Phi'(t)/\Phi(t)$ for $0 \leq t < \rho$, let $\nu = \lim_{t\uparrow\rho} \Psi(t)$, and let $\omega = \sup\{k : w_k > 0\}$. For $0 < \lambda \leq \nu$ let $\tau \in (0, \rho]$ be given by $\Psi(\tau) = \lambda$, with $\Psi(\rho) := \nu$, and let ξ be a random variable with $\mathbb{P}(\xi = k) = w_k\tau^k/\Phi(\tau)$, so that $\mathbb{E}\xi = \lambda$ [2, Sections 3 and 11]. Let $\xi_1, \dots, \xi_n$ be independent copies of ξ and $\xi_{(1)} \geq \dots \geq \xi_{(n)}$ their order statistics, and for $A \subseteq \mathbb{Z}_{\geq 0}$ let

$$N_A = |\{i \leq n : Y_i \in A\}|, \qquad \overline{N}_A = |\{i \leq n : \xi_i \in A\}|.$$

Date: 13 October 2026.

2020 *Mathematics Subject Classification.* 60C05, 60J80, 05C05, 60F05.

Key words and phrases. Conditioned Galton–Watson tree, simply generated tree, random allocation, balls in boxes, maximum degree, infinite variance.

For random variables X and X' with values in $\mathbb{Z}_{\ge 0}$ let

$$d_{\mathrm{K}}(X,X') = \sup_{x\in\mathbb{Z}_{\ge 0}} |\mathbb{P}(X\le x) - \mathbb{P}(X'\le x)|, \qquad d_{\mathrm{TV}}(X,X') = \sup_{A\subseteq\mathbb{Z}_{\ge 0}} |\mathbb{P}(X\in A) - \mathbb{P}(X'\in A)|,$$

and for random variables with values in $\mathbb{Z}_{\ge 0}\cup\{\infty\}$ let

$$\widetilde{d}_{\mathrm{K}}(X,X') = \sup_{x\in\mathbb{Z}_{\ge 0}} \frac{|\mathbb{P}(X\le x) - \mathbb{P}(X'\le x)|}{1+x}.$$

Here $\mathrm{Po}(\infty)$ is the distribution of a random variable that equals ∞ identically, an event holds with high probability (w.h.p.) if its probability tends to 1, and log is the natural logarithm. The following is [2, Theorem 19.7] with the further statements for the case $\lambda<\nu$ omitted.[1]

Theorem A (Janson [2, Theorem 19.7])**.** *Let $w=(w_k)_{k\ge 0}$ be a weight sequence with $w_0>0$ and $\omega=\infty$. Suppose that $n\to\infty$ and $m=m(n)$ with $m=\lambda n+o(\sqrt{n})$ where $0<\lambda\le\nu$. Suppose further that $\sigma^2:=\operatorname{Var}\xi<\infty$. (This is redundant when $\lambda<\nu$.)*

- (i) *If (possibly for n in a subsequence) $h(n)$ are integers such that $n\mathbb{P}(\xi\ge h(n))\to\alpha$, for some $\alpha\in[0,\infty]$, then $N_{[h(n),\infty)}:=|\{i:Y_i\ge h(n)\}|\to\mathrm{Po}(\alpha)$ in distribution.*
- (ii) *If $h(n)$ are integers such that $n\mathbb{P}(\xi\ge h(n))\to 0$, then w.h.p. $Y_{(1)}<h(n)$.*
- (iii) *If $h(n)$ are integers such that $n\mathbb{P}(\xi\ge h(n))\to\infty$, then, for every fixed j, w.h.p. $Y_{(j)}\ge h(n)$.*
- (iv) *For any sequence $h(n)$, $\widetilde{d}_{\mathrm{K}}\bigl(N_{[h(n),\infty)},\overline{N}_{[h(n),\infty)}\bigr)\to 0$.*
- (v) *For every fixed j, $d_{\mathrm{K}}\bigl(Y_{(j)},\xi_{(j)}\bigr)\to 0$.*
- (vi) *$d_{\mathrm{TV}}\bigl(Y_{(1)},\xi_{(1)}\bigr)\to 0$.*

The hypothesis $\sigma^2<\infty$ matters only when $\lambda=\nu$. In [2, Example 19.27], where $\lambda=\nu$ and $w_k\sim ck^{-\alpha-1}$ with $1<\alpha<2$, the asymptotic distributions of $Y_{(1)}$ and $\xi_{(1)}$ are different, while in [2, Example 19.29], where $w_k\sim ck^{-3}$, the conclusions of Theorem A hold although $\sigma^2=\infty$. Janson then wrote: "Note that in Example 19.27, although the asymptotic distributions of $Y_{(1)}$ and $\xi_{(1)}$ are different, they are still of the same order of magnitude. We do not know whether this is true in general. This question can be formulated more precisely as follows."

Problem (Janson [2, Problem 19.30])**.** *In the case $\lambda=\nu$, do Theorem 19.7*(ii)–(iii) *hold also when $\sigma^2=\infty$?*

Kortchemski and Richier [3] studied critical offspring distributions with $\mathbb{P}(\xi\ge k)\sim\ell(k)/k$ as $k\to\infty$, where ℓ is slowly varying; this is the case $\lambda=\nu=1$ with $\sigma^2=\infty$. They proved that the largest outdegree of $\mathcal{T}_n$, divided by $|b_n|$, tends to 1 in probability, while the second, third, ... largest outdegrees, divided by a_n, converge jointly in distribution to the points, in decreasing order, of a Poisson process on $(0,\infty)$ with intensity $x^{-2}\,dx$ [3, Theorem 3]. Here a_n and b_n are explicit sequences with $a_n=o(|b_n|)$, and $\xi_{(1)}/a_n,\xi_{(2)}/a_n,\dots$ converge to the points of the same Poisson process. As they observed, taking $h(n)=(a_n|b_n|)^{1/2}$ shows that part (ii) of Theorem A fails in their setting, which answers Problem 19.30 negatively for part (ii). Their theorem also shows that $Y_{(1)}$ and $\xi_{(1)}$ need not be of the same order of magnitude. Part (iii), on the other hand, holds in their setting: there $n\mathbb{P}(\xi\ge h(n))\to\infty$ forces $h(n)=o(a_n)$, while $Y_{(j)}/a_n$ has a positive limit in distribution for every $j\ge 2$ and $Y_{(1)}/a_n\to\infty$.

We show that part (iii) can fail as well. Our example is in the tree case $m=n-1$, with $\lambda=\nu=1$.

[1] In the arXiv version of [2] the theorem is Theorem 18.7 and the problem below is Problem 18.30.

Theorem 1. *There are a probability distribution w on $\mathbb{Z}_{\ge 0}$ and integers $n_1 < n_2 < \cdots$ and $L_1 < L_2 < \cdots$, all L_r even, with the following properties, where ξ has distribution w and $K_r = (r+\frac{1}{2})L_r$.*

(a) $\mathbb{E}\xi = 1$, $\operatorname{Var}\xi = \infty$, $w_0 > 0$, $w_k > 0$ *for infinitely many* k, *and* $L_r \ge \sqrt{n_r}\,\log n_r$.

(b) $n_r\mathbb{P}(\xi \ge L_r) \ge r$, $n_r\mathbb{P}(\xi \ge K_r) \le \frac{1}{2r+1} + 2^{-r-3}$, *and* $\mathbb{P}(L_r < \xi < K_r) = 0$.

(c) *Let* $n = n_r$ *and* $(Y_1, \dots, Y_n) = B_{n-1,n}$. *Then*

$$\mathbb{P}\Big(Y_{(1)} = K_r \quad \text{and} \quad Y_{(2)} \le \frac{L_r}{64(\log n_r)^2}\Big) \ge 1 - \frac{1}{r}.$$

By the coupling of [2, Lemma 17.1] recalled above, part (c) holds also when $Y_{(1)} \ge Y_{(2)}$ are the two largest outdegrees of the Galton–Watson tree $\mathcal{T}_{n_r}$ with offspring distribution w conditioned to have n_r vertices.

Corollary 2. *Let w, n_r, L_r and K_r be as in Theorem 1, and let $m = n-1$. Then $\rho = \tau = \lambda = \nu = 1$, the random variable ξ of Theorem A has distribution w, and all hypotheses of Theorem A hold except $\sigma^2 < \infty$. Moreover, the following hold, where $n = n_r$ in each probability.*

(a) *Let $h(n_r) = L_r$ for every r and $h(n) = 1$ for all other n. Then $n\mathbb{P}(\xi \ge h(n)) \to \infty$, but $\mathbb{P}(Y_{(2)} \ge h(n_r)) \le 1/r$ and $\mathbb{P}(N_{[h(n_r),\infty)} = 1) \ge 1 - 1/r$.*

(b) *Let $h(n_r) = K_r$ for every r and $h(n) = n$ for all other n. Then $n\mathbb{P}(\xi \ge h(n)) \to 0$, but $\mathbb{P}(Y_{(1)} \ge h(n_r)) \ge 1 - 1/r$ and*

$$\widetilde{d}_{\mathrm{K}}\big(N_{[h(n_r),\infty)}, \overline{N}_{[h(n_r),\infty)}\big) \ge 1 - \frac{1}{r} - \frac{1}{2r+1} - 2^{-r-3}.$$

(c) *For every $j \ge 1$ and every $r \ge 2j$, $d_{\mathrm{K}}(Y_{(j)}, \xi_{(j)}) \ge 1 - 5/r$.*

(d) *As $r \to \infty$,*

$$\mathbb{P}(Y_{(1)} \ge (r+\tfrac{1}{2})L_r) \to 1, \quad \mathbb{P}(\xi_{(1)} \le L_r) \to 1, \quad \mathbb{P}\Big(Y_{(2)} \le \frac{L_r}{64(\log n_r)^2}\Big) \to 1, \quad \mathbb{P}(\xi_{(2)} \ge L_r) \to 1.$$

Consequently each of the conclusions (i)–(vi) *of Theorem A fails:* (i) *with $\alpha = \infty$ by* (a) *and with $\alpha = 0$ by* (b), (ii) *by* (b), (iii) *for every $j \ge 2$ by* (a), (iv) *by* (b), (v) *for every j by* (c), *and* (vi) *by* (c), *since $d_{\mathrm{TV}} \ge d_{\mathrm{K}}$.*

By Theorem 1(c) and part (d), with probability tending to 1 along n_r we have $\xi_{(1)} = \xi_{(2)} = L_r$, $Y_{(1)} = (r+\frac{1}{2})L_r$ and $Y_{(2)} \le L_r/(64(\log n_r)^2)$. So the largest outdegree is larger than $\xi_{(1)}$ by the factor $r + \frac{1}{2} \to \infty$, and the second largest is smaller than $\xi_{(2)}$ by at least the factor $64(\log n_r)^2$.

The distribution w is built in stages. At stage r it has an atom at a frequent value L_r, of probability r/n_r, and an atom at a rare value $K_r = (r+\frac{1}{2})L_r$, of probability $1/((2r+1)n_r)$; every other value is at most $L_r/(64(\log n_r)^2)$ or larger than n_r. Among n_r independent copies of ξ, about r take the value L_r and, with probability about $1/(2r+1)$, one takes the value K_r; on average the two atoms contribute $rL_r + \frac{1}{2}L_r = K_r$ to the sum. If a of the copies take the value L_r and c take the value K_r, then on the event that the sum is $n_r - 1$ the remaining small values must deviate from their mean by $((r+\frac{1}{2})(1-c) - a)L_r$, up to an error of order r. Because of the half in $r + \frac{1}{2}$, the coefficient $(r+\frac{1}{2})(1-c) - a$ is at least $\frac{1}{2}$ in absolute value unless $(a,c) = (0,1)$. A deviation of $L_r/2$ is about $\frac{1}{2}\log n_r$ standard deviations of the sum of the small values, so for $(a,c) \ne (0,1)$ the small values reach the required sum with probability at most $\exp(-c'(\log n_r)^2)$ for a constant $c' > 0$, which is smaller than any power of n_r. The configuration $(a,c) = (0,1)$ has probability of order $e^{-r}/(2r+1)$, and since it needs a deviation of order r only, the small values then reach the required sum with probability of order $n_r^{-1/2}$. For large n_r the conditioning therefore removes every copy of L_r and keeps exactly one copy of K_r.

We do not know whether part (iii) of Theorem A can fail for $j = 1$.

Section 2 gives a local limit estimate with explicit constants. Section 3 constructs w and proves Theorem 1(a) and (b), Section 4 proves Theorem 1(c), and Section 5 proves Corollary 2. Section 6 describes computer checks of the identities and inequalities used in the proofs.

2. A local limit estimate

Lemma 3. *Let $b \geq 1$ be an integer and $0 < \gamma \leq \frac{1}{2}$. Let X be a random variable with values in $\{0, 1, \dots, b\}$ such that $\mathbb{P}(X = 0) \geq \gamma$ and $\mathbb{P}(X = 1) \geq \gamma$, with mean μ and variance v. Let $N \geq 4$, let T_N be the sum of N independent copies of X, and suppose that $s := \sqrt{Nv} \geq 16b$. Then for every integer j,*

$$\Big| s\,\mathbb{P}(T_N = j) - \frac{1}{\sqrt{2\pi}} e^{-(j-N\mu)^2/(2s^2)} \Big| \leq \Big(\frac{b}{s}\Big)^{1/4} + \frac{1}{N} + s\exp\Big(-\frac{2\gamma^2 N}{\pi^2 b^2}\Big).$$

Proof. Let $\varphi(t) = \mathbb{E}e^{it(X-\mu)}$ and $y = (j - N\mu)/s$. By Fourier inversion and the substitution $t = u/s$,

$$s\,\mathbb{P}(T_N = j) = \frac{s}{2\pi}\int_{-\pi}^{\pi} \varphi(t)^N e^{-it(j-N\mu)}\,dt = \frac{1}{2\pi}\int_{|u|\leq \pi s} \varphi(u/s)^N e^{-iuy}\,du,$$

and $(2\pi)^{-1/2}e^{-y^2/2} = (2\pi)^{-1}\int_{\mathbb{R}} e^{-u^2/2}e^{-iuy}\,du$. Let $A = (s/b)^{1/4}$, so that $2 \leq A \leq s/b \leq \pi s$. Then the left side of the claimed inequality is at most $(I_1 + I_2 + I_3 + I_4)/(2\pi)$, where

$$I_1 = \int_{|u|\leq A} |\varphi(u/s)^N - e^{-u^2/2}|\,du, \qquad I_2 = \int_{A<|u|\leq s/b} |\varphi(u/s)|^N\,du,$$
$$I_3 = \int_{s/b<|u|\leq \pi s} |\varphi(u/s)|^N\,du, \qquad I_4 = \int_{|u|>A} e^{-u^2/2}\,du.$$

Since $|X - \mu| \leq b$, we have $v \leq b^2$ and $\mathbb{E}|X - \mu|^3 \leq bv$. Let X' be an independent copy of X; then $|\varphi(t)|^2 = \mathbb{E}\cos(t(X - X'))$ and $\mathbb{E}(X - X')^2 = 2v$.

Let $|u| \leq A$ and $t = u/s$. Then $b|t| \leq bA/s = (b/s)^{3/4} \leq \frac{1}{8}$ and $vt^2 \leq b^2t^2 \leq \frac{1}{64}$. By Taylor's theorem, $|\varphi(t) - 1 + \frac{1}{2}vt^2| \leq \frac{1}{6}bv|t|^3 \leq \frac{1}{48}vt^2$, so $|\varphi(t)| \leq 1 - \frac{23}{48}vt^2 \leq e^{-23vt^2/48}$. Also $|e^{-vt^2/2} - 1 + \frac{1}{2}vt^2| \leq \frac{1}{8}v^2t^4$ and $e^{-vt^2/2} \leq e^{-23vt^2/48}$. Using $|z^N - z'^N| \leq N|z - z'|\max(|z|, |z'|)^{N-1}$, $N - 1 \geq \frac{3}{4}N$ and $Nvt^2 = u^2$, we get

$$|\varphi(t)^N - e^{-u^2/2}| \leq N\Big(\frac{bv|t|^3}{6} + \frac{v^2t^4}{8}\Big)e^{-\frac{23}{64}u^2} = \Big(\frac{b|u|^3}{6s} + \frac{u^4}{8N}\Big)e^{-\alpha u^2}, \qquad \alpha = \frac{23}{64}.$$

Since $\int_{\mathbb{R}} |u|^3 e^{-\alpha u^2}\,du = \alpha^{-2}$ and $\int_{\mathbb{R}} u^4 e^{-\alpha u^2}\,du = \frac{3}{4}\sqrt{\pi}\,\alpha^{-5/2}$, this gives $I_1 \leq 1.2905\,b/s + 2.1463/N$.

For $|t| \leq 1/b$ we have $|t(X - X')| \leq 1$, and $1 - \cos x \geq \frac{x^2}{2} - \frac{x^4}{24} \geq \frac{11}{24}x^2$ for $|x| \leq 1$, so $|\varphi(t)|^2 \leq 1 - \frac{11}{24}t^2\mathbb{E}(X - X')^2 = 1 - \frac{11}{12}vt^2 \leq e^{-11vt^2/12}$. Hence $|\varphi(u/s)|^N \leq e^{-11u^2/24}$ for $|u| \leq s/b$. Using $\int_{|u|>A} e^{-\beta u^2}\,du \leq e^{-\beta A^2}/(\beta A)$ for $\beta > 0$, and $A \geq 2$,

$$I_2 \leq \frac{24}{11}\cdot\frac{e^{-11/6}}{A}, \qquad I_4 \leq \frac{2e^{-2}}{A}.$$

For $1/b < |t| \leq \pi$, since $1 - \cos t \geq 2t^2/\pi^2$ for $|t| \leq \pi$ (the function $(1 - \cos t)/t^2$ decreases on $(0, \pi]$) and $\mathbb{P}(|X - X'| = 1) \geq 2\mathbb{P}(X = 0)\mathbb{P}(X = 1) \geq 2\gamma^2$,

$$1 - |\varphi(t)|^2 = \mathbb{E}(1 - \cos(t(X - X'))) \geq \mathbb{P}(|X - X'| = 1)(1 - \cos t) \geq \frac{4\gamma^2t^2}{\pi^2} \geq \frac{4\gamma^2}{\pi^2 b^2}.$$

Hence $|\varphi(t)|^N \leq \exp(-2\gamma^2 N/(\pi^2 b^2))$ there, and $I_3 \leq 2\pi s\exp(-2\gamma^2 N/(\pi^2 b^2))$.

Finally $b/s = (b/s)^{1/4}(b/s)^{3/4} \le \frac18 (b/s)^{1/4}$ and $1/A = (b/s)^{1/4}$, so

$$\frac{I_1+I_2+I_3+I_4}{2\pi} \le \frac{1}{2\pi}\Big(\frac{1.2905}{8} + \frac{24}{11}e^{-11/6} + 2e^{-2}\Big)\Big(\frac{b}{s}\Big)^{1/4} + \frac{2.1463}{2\pi N} + s\exp\Big(-\frac{2\gamma^2 N}{\pi^2 b^2}\Big).$$

The two constants are less than 0.125 and 0.342, which proves the lemma. □

3. The construction

The distribution w is built from integers n_r and even integers L_r, $r \ge 1$, chosen below. Put $K_r = (r+\frac12)L_r$ and

$$p_r = \frac{r}{n_r}, \qquad q_r = \frac{1}{(2r+1)n_r}, \qquad \eta_r = p_r L_r + q_r K_r = \frac{K_r}{n_r}, \tag{3.1}$$

and let $\kappa_1 = \sum_{r\ge1}(p_r+q_r)$ and $\kappa_2 = \sum_{r\ge1}\eta_r$. Provided that $\kappa_1, \kappa_2 \le \frac18$ and that the numbers $0, 1, 2, L_1, K_1, L_2, K_2, \dots$ are distinct, let w be the distribution with

$$w_{L_r} = p_r, \quad w_{K_r} = q_r \quad (r\ge1), \qquad w_0 = \tfrac14 - \kappa_1 + \kappa_2, \quad w_1 = \tfrac12 - \kappa_2, \quad w_2 = \tfrac14, \tag{3.2}$$

and $w_k = 0$ for all other k. Its total mass is 1 and its mean is $w_1 + 2w_2 + \kappa_2 = 1$.

The integers are chosen recursively. Let $n_0 = 1$ and $K_0 = 2$. Let $r \ge 1$, suppose that n_i and L_i have been chosen for $1 \le i < r$, and put

$$V_r = \frac32 + \sum_{i=1}^{r-1}(p_i L_i^2 + q_i K_i^2), \qquad b_r = K_{r-1}.$$

For an integer $n \ge 2$ let $s = \sqrt{nV_r}$ and $\Lambda = \log n$, and let L be the least even integer with $L \ge \Lambda s$. We let $n_r = n$ and $L_r = L$ for an n that satisfies the following five conditions, where $b = b_r$ and $V = V_r$.

(C1) $n \ge 2^{r+3}(r+1)\,n_{r-1}$ and $(r+\frac12)L \le 2^{-r-3}\,n/n_{r-1}$.
(C2) $L > 2\max(b, n_{r-1})$ and $L \ge 13r + 8$.
(C3) $s \ge 64 b\Lambda$ and $n \ge 800(r+1)^2 + 1$.
(C4) $(n-1)V \ge 6\cdot 40^8\, b^2$ and $2s\exp(-(n-1)/(32\pi^2 b^2)) \le \frac{1}{40}$.
(C5) $7(2r+1)e^{r+1} s\, e^{-\Lambda^2/37} \le \frac1r$.

Such an n exists. Indeed, b, V and n_{r-1} do not depend on n, while s is of order $\sqrt n$ and L of order $\sqrt n \log n$; hence each condition holds for all large n, for (C5) because $s\,e^{-\Lambda^2/37} = \sqrt{nV}\,e^{-(\log n)^2/37} \to 0$. For example, the least powers of 2 that satisfy (C1)–(C5) at the first five stages are $n_1 = 2^{47}$, $n_2 = 2^{132}$, $n_3 = 2^{322}$, $n_4 = 2^{725}$ and $n_5 = 2^{1556}$ (Section 6).

Lemma 4. (a) *$2 < L_1$, and $b_r < L_r < K_r < n_r < L_{r+1}$ for every $r \ge 1$. In particular $0 < 1 < 2 < L_1 < K_1 < L_2 < K_2 < \cdots$.*
(b) *For every $i \ge 1$, $p_i + q_i \le 2^{-i-3}/n_{i-1}$ and $\eta_i \le 2^{-i-3}/n_{i-1}$. Hence $\kappa_1, \kappa_2 \le \frac18$, and w is a probability distribution with mean 1, $w_0 \ge \frac18$ and $w_1 \ge \frac38$.*
(c) *$p_r L_r^2 \ge r(\log n_r)^2 V_r$ for every r. In particular $\operatorname{Var}\xi = \infty$.*
(d) *Let $\zeta_r = \sum_{i>r}(p_i+q_i)$ and $\zeta'_r = \sum_{i>r}\eta_i$. Then $n_r\zeta_r \le 2^{-r-3}$ and $n_r\zeta'_r \le 2^{-r-3}$.*
(e) *Theorem 1*(a) *and* (b) *hold.*

Proof. (a) By (C2), $L_1 > 2\max(b_1, n_0) = 4$ and $b_r < L_r$; by (C1), $K_r = (r+\frac12)L_r \le 2^{-r-3}n_r/n_{r-1} < n_r$; and by (C2) at stage $r+1$, $L_{r+1} > 2n_r$. Since $b_{r+1} = K_r$, the chain follows.

(b) By (C1), $p_i + q_i \le (i+1)/n_i \le 2^{-i-3}/n_{i-1}$ and $\eta_i = (i+\frac12)L_i/n_i \le 2^{-i-3}/n_{i-1}$. As $n_{i-1} \ge 1$, it follows that $\kappa_1, \kappa_2 \le \sum_{i\ge1} 2^{-i-3} = \frac18$. Together with (a), this shows that (3.2)

defines a probability distribution, with $w_0 \ge \frac14 - \frac18$ and $w_1 \ge \frac12 - \frac18$; its mean is 1 as computed above.

(c) Since $L_r \ge \Lambda s$ with $\Lambda = \log n_r$ and $s^2 = n_r V_r$, we have $p_r L_r^2 \ge (r/n_r)(\log n_r)^2 n_r V_r = r(\log n_r)^2 V_r$. As $V_r \ge \frac32$ and $n_r \ge 3$, $\mathbb{E}\xi^2 \ge \sum_r p_r L_r^2 = \infty$.

(d) For $i > r$ we have $n_{i-1} \ge n_r$, so by (b) $n_r\zeta_r \le \sum_{i>r} 2^{-i-3} = 2^{-r-3}$, and likewise $n_r\zeta'_r \le 2^{-r-3}$.

(e) By (a), $L_r < n_r < L_{r+1} < n_{r+1}$, so both sequences increase, and every L_r is even by construction. The first four statements of Theorem 1(a) follow from (b) and (c), and $L_r \ge \log n_r \sqrt{n_r V_r} \ge \sqrt{n_r}\log n_r$. By (a), the values of ξ that are at least L_r are L_r, K_r and the values L_i, K_i with $i > r$, and no value lies strictly between L_r and K_r. Hence $n_r\mathbb{P}(\xi \ge L_r) \ge n_r(p_r + q_r) \ge r$ and, by (d), $n_r\mathbb{P}(\xi \ge K_r) = n_r(q_r + \zeta_r) \le \frac{1}{2r+1} + 2^{-r-3}$. □

4. Proof of Theorem 1(c)

Fix $r \ge 1$ and write $n = n_r$, $m = n-1$, $L = L_r$, $K = K_r$, $p = p_r$, $q = q_r$, $b = b_r$, $V = V_r$, $s = \sqrt{nV}$ and $\Lambda = \log n$. Let $\xi_1, \dots, \xi_n$ be independent with distribution w and $S_n = \xi_1 + \dots + \xi_n$, so that $B_{m,n}$ is $(\xi_1, \dots, \xi_n)$ conditioned on $S_n = m$; here $\mathbb{P}(S_n = m) > 0$ since $w_0, w_1 > 0$. By Lemma 4(a), every value of ξ is either *low*, in $[0, b]$, or equal to L, or equal to K, or *high*, at least L_{r+1}, and $L_{r+1} > n > m$. Let $\theta = \mathbb{P}(\xi \le b)$, let X be a random variable with the distribution of ξ conditioned on $\xi \le b$, with mean μ and variance v, and let T_N be the sum of N independent copies of X, with $T_0 = 0$.

Lemma 5. *We have $\theta \ge \frac78$, $\mu \le \frac87$, $\mathbb{P}(X = 0) \ge \frac18$, $\mathbb{P}(X = 1) \ge \frac38$, $\frac16 V \le v \le \frac87 V$, and $b \le s/(64\Lambda) \le L/(64\Lambda^2)$.*

Proof. We have $\theta = 1 - p - q - \zeta_r \ge 1 - \kappa_1 \ge \frac78$, $\mu = \mathbb{E}[\xi; \xi \le b]/\theta \le \mathbb{E}\xi/\theta \le \frac87$, $\mathbb{P}(X = 0) = w_0/\theta \ge w_0 \ge \frac18$ and $\mathbb{P}(X = 1) \ge w_1 \ge \frac38$. The low values are 0, 1, 2 and L_i, K_i for $i < r$, so $\mathbb{E}[\xi^2; \xi \le b] = w_1 + 4w_2 + \sum_{i<r}(p_i L_i^2 + q_i K_i^2) = V - \kappa_2$. Hence $v \le \mathbb{E}X^2 \le V/\theta \le \frac87 V$. If $r \ge 2$, then $V \ge \frac32 + p_1 L_1^2 \ge 2$ by Lemma 4(c), and $v = \mathbb{E}X^2 - \mu^2 \ge V - \kappa_2 - (\frac87)^2 \ge V - \frac18 - \frac{64}{49} \ge \frac16 V$. If $r = 1$, then $b = 2$, $V = \frac32$, and X takes the values $0, 1, 2$ with probabilities $x_0 \ge \frac18$, $x_1 \ge \frac38$ and $x_2 = w_2/\theta \ge \frac14$. The variance $x_0x_1 + 4x_0x_2 + x_1x_2$ is a concave function of (x_0, x_1, x_2) on the simplex, so on the triangle $\{x_0 \ge \frac18,\ x_1 \ge \frac38,\ x_2 \ge \frac14,\ x_0 + x_1 + x_2 = 1\}$ its minimum is attained at a vertex. The values at the vertices $(\frac38, \frac38, \frac14)$, $(\frac18, \frac58, \frac14)$ and $(\frac18, \frac38, \frac12)$ are $\frac{39}{64}$, $\frac{23}{64}$ and $\frac{31}{64}$, so $v \ge \frac{23}{64} > \frac14 = \frac16 V$. Finally, $b \le s/(64\Lambda)$ by (C3), and $s \le L/\Lambda$ by the choice of L. □

For integers $a, c \ge 0$ let $E_{a,c}$ be the event that exactly a of $\xi_1, \dots, \xi_n$ equal L and exactly c of them equal K. On $\{S_n = m\}$ no ξ_i is high, so on $E_{a,c} \cap \{S_n = m\}$ the other $n - a - c$ variables are low and their sum is $m - aL - cK$. Hence, for $a + c \le n$,

$$\mathbb{P}(S_n = m,\, E_{a,c}) = \binom{n}{a,\, c,\, n-a-c} p^a q^c \theta^{n-a-c}\, \mathbb{P}(T_{n-a-c} = m - aL - cK), \tag{4.1}$$

and $\mathbb{P}(S_n = m, E_{a,c}) = 0$ for $a + c > n$. On $E_{0,1} \cap \{S_n = m\}$ we have $Y_{(1)} = K$ and $Y_{(2)} \le b \le L/(64\Lambda^2)$ by Lemma 5. Since $(1 + 1/r)^{-1} \ge 1 - 1/r$, Theorem 1(c) follows from

$$\sum_{(a,c)\neq(0,1)} \mathbb{P}(S_n = m,\, E_{a,c}) \le \frac{1}{r}\,\mathbb{P}(S_n = m,\, E_{0,1}). \tag{4.2}$$

The deviations. Let $x = n\zeta_r$ and $y = n\zeta'_r$, so that $0 \le x, y \le 2^{-r-3} \le \frac{1}{16}$ by Lemma 4(d). Splitting $\mathbb{E}\xi = 1$ according to the classes of values gives $\theta\mu + \eta_r + \zeta'_r = 1$, that is, $n\theta\mu = n - K - y$;

also $n\theta = n - r - \frac{1}{2r+1} - x$. For $a + c \le n$ let

$$D(a,c) = m - aL - cK - (n-a-c)\mu$$

be the deviation of $m - aL - cK$ from the mean of T_{n-a-c}. Writing $m = n\theta\mu + K + y - 1$ and $K = (r+\frac{1}{2})L$, a direct computation gives

$$D(a,c) = \big((r+\tfrac{1}{2})(1-c) - a\big)L + (a+c-1)\mu + D_0, \tag{4.3}$$

where

$$D_0 := D(0,1) = \Big(1 - r - \frac{1}{2r+1} - x\Big)\mu + y - 1.$$

The coefficient of μ in D_0 lies in $[-(r - \frac{2}{3} + \frac{1}{16}), 0]$ and $0 \le \mu \le \frac{8}{7}$, so

$$|D_0| \le \frac{8}{7}\Big(r - \frac{2}{3} + \frac{1}{16}\Big) + 1 \le 2r + 2. \tag{4.4}$$

Lemma 6. *Let $r \ge 1$, $L \ge 13r + 8$, $0 \le \mu \le \frac{8}{7}$ and $|D_0| \le 2r + 2$. Then for all integers $a, c \ge 0$ with $(a,c) \ne (0,1)$,*

$$\Delta := \big((r+\tfrac{1}{2})(1-c) - a\big)L + (a+c-1)\mu + D_0 \quad \textit{satisfies} \quad |\Delta| \ge \frac{L}{4}.$$

Proof. Note that $L - \mu \ge 0$. If $c = 0$ and $a \le r$, then $\Delta = (r + \frac{1}{2} - a)L + (a-1)\mu + D_0 \ge \frac{1}{2}L - \frac{8}{7} - 2r - 2 \ge \frac{1}{4}L$, since $L \ge 13r + 8 \ge 8r + \frac{88}{7}$. If $c = 0$ and $a \ge r + 1$, then

$$\Delta = -(a - r - \tfrac{1}{2})(L - \mu) + (r - \tfrac{1}{2})\mu + D_0 \le -\tfrac{1}{2}(L-\mu) + \tfrac{8}{7}(r - \tfrac{1}{2}) + 2r + 2 \le -\tfrac{1}{2}L + \tfrac{22}{7}r + 2 \le -\tfrac{1}{4}L,$$

since $L \ge 13r + 8 \ge \frac{88}{7}r + 8$. If $c = 1$, then $a \ge 1$ and $\Delta = -a(L-\mu) + D_0 \le -L + \frac{8}{7} + 2r + 2 \le -\frac{1}{4}L$. If $c \ge 2$, then

$$\Delta = -(c-1)\big((r+\tfrac{1}{2})L - \mu\big) - a(L-\mu) + D_0 \le -(r+\tfrac{1}{2})L + \tfrac{8}{7} + 2r + 2 \le -\tfrac{1}{4}L. \qquad \square$$

The configurations other than $(0,1)$**.** Let $(a,c) \ne (0,1)$ with $a + c \le n$, and put $N = n - a - c$ and $j = m - aL - cK$, so that $j - N\mu = D(a,c)$. By (C2), Lemma 5, (4.3), (4.4) and Lemma 6, $|j - N\mu| \ge L/4 \ge \Lambda s/4$. If $N = 0$, then $|j| \ge L/4 > 0$ and $\mathbb{P}(T_0 = j) = 0$. If $N \ge 1$, then, since $|X - \mu| \le b$ and $Nv \le nv \le \frac{8}{7}s^2$, Bernstein's inequality [1] with $t = \Lambda s/4$ gives

$$\mathbb{P}(T_N = j) \le \mathbb{P}(|T_N - N\mu| \ge t) \le 2\exp\Big(-\frac{t^2}{2(Nv + bt/3)}\Big) \le 2\exp\Big(-\frac{\Lambda^2}{32(\frac{8}{7} + \frac{1}{768})}\Big) \le 2e^{-\Lambda^2/37},$$

where we used $bt/3 = b\Lambda s/12 \le s^2/768$, by Lemma 5. By (4.1) and the multinomial theorem,

$$\sum_{(a,c)\ne(0,1)} \mathbb{P}(S_n = m,\, E_{a,c}) \le 2e^{-\Lambda^2/37} \sum_{a+c\le n} \binom{n}{a,\, c,\, n-a-c} p^a q^c \theta^{n-a-c} \le 2e^{-\Lambda^2/37}. \tag{4.5}$$

The configuration $(0,1)$**.** By (4.1), $\mathbb{P}(S_n = m, E_{0,1}) = nq\,\theta^{n-1}\,\mathbb{P}(T_{n-1} = m - K)$, and $nq = 1/(2r+1)$. Write $\theta = 1 - z$, where $nz = n(p + q + \zeta_r) = r + \frac{1}{2r+1} + x \le r + \frac{1}{3} + \frac{1}{16}$. Then $z \le \frac{1}{2}$ and $nz^2 \le (r+1)^2/n \le \frac{1}{2}$ by (C3), and since $1 - z \ge e^{-z-z^2}$ for $0 \le z \le \frac{1}{2}$,

$$\theta^{n-1} \ge \theta^n \ge e^{-nz - nz^2} \ge e^{-r - \frac{1}{3} - \frac{1}{16} - \frac{1}{2}} \ge e^{-r-1}.$$

We apply Lemma 3 with $\gamma = \frac{1}{8}$, $N = n - 1$ and $j = m - K$, for which $j - N\mu = D_0$. Let $s' = \sqrt{(n-1)v}$. By Lemma 5 and (C4), $s'^2 \ge (n-1)V/6 \ge 40^8 b^2$, so $s' \ge 16b$ and $(b/s')^{1/4} \le \frac{1}{40}$. By (C3), $1/N \le \frac{1}{40}$. Since $s' \le \sqrt{8/7}\,s \le 2s$ and $2\gamma^2 N/(\pi^2 b^2) = (n-1)/(32\pi^2 b^2)$, (C4) gives

$s' \exp(-2\gamma^2 N/(\pi^2 b^2)) \le \frac{1}{40}$. Moreover $s'^2 \ge (n-1)V/6 \ge (n-1)/4 \ge 200(r+1)^2$ by (C3), so $D_0^2/(2s'^2) \le (2r+2)^2/(400(r+1)^2) = 0.01$ by (4.4). Lemma 3 therefore gives

$$s'\,\mathbb{P}(T_{n-1} = m-K) \ge \frac{e^{-0.01}}{\sqrt{2\pi}} - \frac{3}{40} > 0.3199,$$

so $\mathbb{P}(T_{n-1} = m-K) > 0.3199/(\sqrt{8/7}\,s) > 0.29/s$, and

$$\mathbb{P}(S_n = m,\, E_{0,1}) \ge \frac{e^{-r-1}}{2r+1} \cdot \frac{0.29}{s}.$$

Together with (4.5) and (C5),

$$\frac{\sum_{(a,c)\neq(0,1)} \mathbb{P}(S_n = m, E_{a,c})}{\mathbb{P}(S_n = m, E_{0,1})} \le \frac{2}{0.29}(2r+1)e^{r+1}s\,e^{-\Lambda^2/37} \le 7(2r+1)e^{r+1}s\,e^{-\Lambda^2/37} \le \frac{1}{r},$$

which is (4.2). This proves Theorem 1(c).

5. Proof of Corollary 2

Since $\mathbb{E}\xi = 1$, the radius of convergence ρ of $\Phi(t) = \mathbb{E}t^\xi$ is at least 1, and since $\mathbb{E}\xi^2 = \infty$ it is at most 1. As $t \uparrow 1$, $\Phi(t) \to 1$ and $\Phi'(t) \to \mathbb{E}\xi = 1$ by monotone convergence, so $\nu = \lim_{t\uparrow 1} \Psi(t) = 1$. For $m = n-1 = n + o(\sqrt{n})$ we have $\lambda = 1 = \nu$ and $\tau = \rho = 1$, and the random variable ξ of Theorem A has $\mathbb{P}(\xi = k) = w_k\tau^k/\Phi(\tau) = w_k$. By Theorem 1(a), $w_0 > 0$, $\omega = \infty$ and $\sigma^2 = \infty$.

Let $\mathcal{G}_r$ be the event in Theorem 1(c), for $n = n_r$. On $\mathcal{G}_r$ exactly one Y_i is at least L_r, and it equals K_r, since $L_r/(64(\log n_r)^2) < L_r$. Let $\bar\mu_r = n_r\mathbb{P}(\xi \ge L_r) \ge r$. For $n = n_r$ the variable $\overline{N}_{[L_r,\infty)}$ has the binomial distribution with mean $\bar\mu_r$ and variance at most $\bar\mu_r$, so for every $j \ge 1$ and $r \ge 2j$ we have $j - 1 < \bar\mu_r/2$ and, by Chebyshev's inequality,

$$\mathbb{P}(\overline{N}_{[L_r,\infty)} \le j-1) \le \mathbb{P}(|\overline{N}_{[L_r,\infty)} - \bar\mu_r| \ge \bar\mu_r/2) \le \frac{4}{\bar\mu_r} \le \frac{4}{r}. \tag{5.1}$$

Also $\mathbb{P}(\overline{N}_{[K_r,\infty)} = 0) \ge 1 - n_r\mathbb{P}(\xi \ge K_r) \ge 1 - \frac{1}{2r+1} - 2^{-r-3}$ by Theorem 1(b).

(a) For $n = n_r$, $n\mathbb{P}(\xi \ge h(n)) \ge r$, and for other n, $n\mathbb{P}(\xi \ge 1) = n(1-w_0) \ge \frac{5}{8}n$. On $\mathcal{G}_r$ we have $Y_{(2)} < L_r$ and $N_{[L_r,\infty)} = 1$, which gives the two bounds.

(b) For $n = n_r$, $n\mathbb{P}(\xi \ge h(n)) \le \frac{1}{2r+1} + 2^{-r-3}$, and for other n, $n\mathbb{P}(\xi \ge n) \le \mathbb{E}[\xi; \xi \ge n] \to 0$. On $\mathcal{G}_r$ we have $Y_{(1)} = K_r$ and $N_{[K_r,\infty)} = 1$, so $\mathbb{P}(Y_{(1)} \ge K_r) \ge 1 - \frac{1}{r}$ and, taking $x = 0$ in the definition of $\widetilde{d}_{\mathrm{K}}$,

$$\widetilde{d}_{\mathrm{K}}\big(N_{[K_r,\infty)}, \overline{N}_{[K_r,\infty)}\big) \ge \mathbb{P}\big(\overline{N}_{[K_r,\infty)} = 0\big) - \mathbb{P}\big(N_{[K_r,\infty)} = 0\big) \ge 1 - \frac{1}{2r+1} - 2^{-r-3} - \frac{1}{r}.$$

(c) For $j = 1$, $\mathbb{P}(Y_{(1)} \le K_r - 1) \le \mathbb{P}(\mathcal{G}_r^c) \le \frac{1}{r}$ and $\mathbb{P}(\xi_{(1)} \le K_r - 1) = \mathbb{P}(\overline{N}_{[K_r,\infty)} = 0) \ge 1 - \frac{1}{2r+1} - 2^{-r-3}$, so $d_{\mathrm{K}}(Y_{(1)}, \xi_{(1)}) \ge 1 - \frac{3}{r}$. For $j \ge 2$, $\mathbb{P}(Y_{(j)} \le L_r - 1) \ge \mathbb{P}(\mathcal{G}_r) \ge 1 - \frac{1}{r}$, while $\mathbb{P}(\xi_{(j)} \le L_r - 1) = \mathbb{P}(\overline{N}_{[L_r,\infty)} \le j-1) \le \frac{4}{r}$ by (5.1); so $d_{\mathrm{K}}(Y_{(j)}, \xi_{(j)}) \ge 1 - \frac{5}{r}$.

(d) The first and third limits hold because $\mathbb{P}(\mathcal{G}_r) \ge 1 - \frac{1}{r}$. Since no value of ξ lies strictly between L_r and K_r, $\mathbb{P}(\xi_{(1)} \le L_r) = \mathbb{P}(\overline{N}_{[K_r,\infty)} = 0) \to 1$. By (5.1) with $j = 2$, $\mathbb{P}(\xi_{(2)} \ge L_r) = \mathbb{P}(\overline{N}_{[L_r,\infty)} \ge 2) \ge 1 - \frac{4}{r}$ for $r \ge 4$.

For the final statement, (i) with $\alpha = \infty$ would give $N_{[h(n),\infty)} \to \infty$ in probability, and with $\alpha = 0$ it would give $N_{[h(n),\infty)} \to 0$ in probability; (a) and (b) exclude both. The remaining failures are immediate from (a)–(c), using $d_{\mathrm{TV}} \ge d_{\mathrm{K}}$ [2, Lemma 19.5] for (vi). □

6. Computations

The repository listed under Data availability contains a Python script, `checks.py`, with its output, and a Lean file, `CaseBound.lean`. The script computes the conditional distribution of (a, c) given $S_n = n - 1$ for a one-stage version of w, with values $0, 1, 2, L, K$, $p = r/n$ and $q = 1/((2r + 1)n)$, from (4.1) in double precision, with a pruned weight below 10^{-20} of the total, and checks (4.1) against a direct convolution in three small cases. For $n = 10^6$ and L the least even integer at least $s \log n$, the conditional probability that $(a, c) \neq (0, 1)$ is $2.9 \cdot 10^{-28}$ for $r = 2$ and $9.7 \cdot 10^{-26}$ for $r = 4$, while for independent copies the probability that at least two of them are at least L is 0.645 and 0.916. When L is instead about s, the conditional probability that at least two values are at least L is about 0.90 for $r = 2$ and 0.99 for $r = 4$. The script also checks (4.3) in exact rational arithmetic, Lemma 6 on 346,500 cases, together with a control showing that the bound fails at a vertex for every even L just below $\frac{88}{7}r + 8$, $r \leq 30$; Lemma 3 against the distributions of T_N, computed by the fast Fourier transform, for 48 random laws, where the largest ratio of the error to the bound is 0.0015, together with a control showing that without the atom at 1 the error stays near $(2\pi)^{-1/2}$; the numerical constants in Sections 2–4; and conditions (C1)–(C5) for $r \leq 5$ in high-precision arithmetic. The Lean file, checked with Lean 4 [4] (version 4.33.1) and Mathlib [5], proves (4.3), the bound $|D_0| \leq 2r + 2$ of (4.4), and Lemma 6.

These computations test the identities and inequalities used in the proofs; they are not part of them.

Acknowledgements and funding

The research presented in this paper was supported by the European Research Council (ERC) under the European Union's Horizon Europe research and innovation programme (grant agreement No. 101041711), by the Simons Foundation as part of the Collaboration on the Mathematical and Scientific Foundations of Deep Learning, by Heights Labs, by Convex Nexus Capital, by the Israel Science Foundation (grant number 2258/19), and by the Israel Science Foundation (ISF Grant 4101/25). Computational resources were provided in part by Google Cloud through its education credits program (credit 530550103).

This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [6] and working together inside the Brain framework [8], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [7], the Math Brain repository [6] and the Brain framework repository [8].

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. Problem 19.30 of [2] was among the open problems in probability on which the project's agents worked in September 2026. On 14 September 2026 Claude Fable 5.1 (Anthropic) found the mechanism behind Theorem 1, a rare value that exceeds the mass missing from the sum by exactly the expected total of a frequent value, so that the conditioning removes every copy of the frequent value; it sketched a proof and checked the mechanism on an exact example. On 13 October 2026 Claude Opus 5.5 (Anthropic), working in Claude Code, found the construction of Section 3 and the proofs in their present form, including Lemma 3, searched the literature, and

wrote the computer checks, the Lean file and the manuscript. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No data were used. The code and the Lean file are available at `https://github.com/yspennstate/second-largest-degree-infinite-variance`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [6].

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